

be scrapped or placed into lower yield categories, wasting valuable production time and losing potential profit.

Fig. 3 and Table II illustrate this point. These show printing techniques for a standard single-printing process on 15.2cm (six-inch) multicrystalline wafers in six categories of sorting (excluding rejected cells). Better print quality results in a percentage of total yearly cells moving from bottom to top categories.

As these examples of breakage and print alignment show, there is quantifiable financial benefit in looking beyond the cost of equipment and understanding the value that can be gained by process and yield improvement. Scaling production is important in reducing manufacturing costs. Future cost improvements will require major changes in technology and manufacturing methods to improve efficiency at high yield and very high factory outputs. Driving down the cost per watt even further, with the speed required to quickly expand global grid parity, calls for process control and optimisation.

Calculating value

Table III shows several calculations that quantify benefits not normally

TABLE II
Scenarios of X% of Cells Moving to Higher Categories

(Assuming solar cell ASP [\$]/Wp) of \$0.86 for 90MW line)

	Baseline	1%	3%	5%
Annual cells produced	22-24 million			
Profit increase (over 5 years)	—	\$160,000	\$500,000	\$800,000

TABLE III
Estimated Dollar Benefit of An Innovative Platform

Area of benefit	Description	Expected benefit over 5 years vs industry standard
Platform: High mechanical yield	Profit gain on unbroken wafers	\$1,628,650
Platform: Easy addition of process step	Heavy reconfiguration once in 5 years, such as an upgrade to double printing	\$46,800
Platform: Single-wire concept	Savings on IT projects	\$37,800
Platform: Extendability	Longer amortisation (actual)	\$800,000
Printer: Consumables savings	Extended lifetime of print screens	\$33,333
Printer: Better process yield	Extra profit	\$84,240
Total expected benefit per tool (in addition to TCO)		\$2,630,823

captured by a TCO spreadsheet. The table shows, for example, that improvements in process yield and reducing direct consumables costs can bring a competitive advantage greater than \$100,000 through newer and innovative screen-printing metallisation. Overall, benefits such as easy integration of new process steps, an extendable platform and con-

sumables cost savings could result in about a \$2-million advantage over five years versus a traditional cell manufacturing platform that is not highly innovative.

To sum up

Challenges and opportunities from newer cell structures, adoption of newer and additional processes, and a higher sensitivity to process optimisation suggest that evaluating TCO is no longer sufficient to assess the quality of a manufacturing tool which is a great investment. Such an investment should be evaluated in the light of its potential for generating differential value along the years.

There are a number of areas where evaluating a tool solely on TCO does not capture the tool's value. In fact, quantifying some of these values shows that the added value of the tool may even be comparable with the tool procurement cost itself.

Significant innovation will lead to a market with new concepts that help accelerate the \$/Wp reduction towards and beyond grid parity. ●

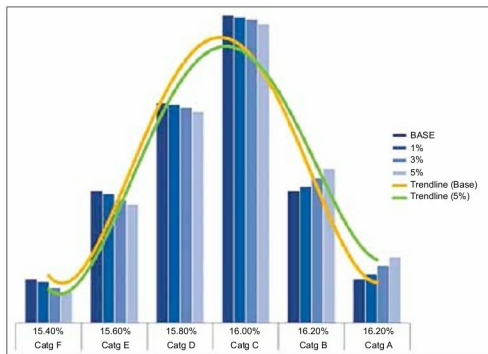


Fig. 3: Model illustrating how better print quality results in more wafers in the higher efficiency categories